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QUALITY OF LIFE INDEX FOR PORTUGUESE MUNICIPALITIES

UNSUPERVISED STATISTICAL LEARNING

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1. Introduction

Quality of life in territories is a multidimensional concept that integrates economic, social, demographic, environmental and institutional factors, playing a central role in the analysis of sustainable development and in the design of public policies. The existence of significant structural differences between municipalities makes it particularly relevant to build analytical instruments that make it possible to synthesise this diversity and to support territorial comparison in a consistent and informed way.

The aim of this work is to build a municipal quality of life index, using unsupervised statistical learning techniques, within the scope of a decision-support exercise requested by a Danish non-governmental organisation dedicated to the comparative study of quality of life in the countries of the Mediterranean region. In this context, the study carried out corresponds to the application of the model to the Portuguese case, serving as a national-scale analysis and as a methodological basis for future extensions to other countries.

The analysis is based on official statistical information made available by the Instituto Nacional de Estatística (Statistics Portugal), covering a broad set of indicators that reflect different dimensions of municipal quality of life, including economic, social, demographic, territorial, cultural and environmental conditions. Through the identification of latent dimensions underlying the data and the subsequent classification of municipalities into homogeneous groups, the study seeks to capture common structural patterns, without imposing predefined classifications or normative judgements.

The substantive relevance of this work lies in its ability to transform a large volume of scattered statistical information into a synthetic index and a typology of municipalities, facilitating a comparative reading of the Portuguese territory. The results obtained are not intended to establish an absolute ranking of municipalities, but rather to provide an analytical instrument that makes it possible to understand different quality of life profiles and to support the strategic reflection of entities involved in territorial analysis and in the promotion of the well-being of populations.

2. Data

2.1 Summary description of the database

The database used in this study consists exclusively of variables taken from the online platform of the Instituto Nacional de Estatística, ensuring the consistency, reliability and comparability of the information used. The unit of analysis is the municipality, covering all Portuguese municipalities of Mainland Portugal, the Autonomous Region of the Azores and the Autonomous Region of Madeira, which allows a comprehensive analysis of quality of life at the national scale. The information collected was aggregated into a single database, corresponding to the file “dataset_completo.csv”, which is the starting point for all subsequent stages of the analysis.

Before building the final dataset, it was necessary to carry out data preparation work, since the files made available by INE in CSV format contain, by default, more information than is strictly necessary for the present analysis. In this context, the variables were selected, harmonised and normalised so as to ensure their comparability across municipalities. The initial set comprised 40 candidate INPUT variables, of which 30 active variables were subsequently used in the analysis, complemented by a set of passive variables (PROFILE) intended exclusively for the characterisation and interpretation of the results.

2.2 Variables

The active variables presented below correspond to the initial set of indicators considered for the analysis. The selection of the final subset of variables used in identifying the latent dimensions is discussed in Section 3.
Active variables (INPUT):

1. Bathing waters (No.) by Geographic location
2. Beneficiaries of the social integration income (rendimento social de inserção) of social security per 1000 working-age inhabitants (‰)
3. Immovable cultural properties (No.) by Geographic location
4. Firefighters (No.) by Geographic location
5. Gini coefficient of declared gross income minus assessed personal income tax (IRS) per taxpayer (%) by Geographic location
6. Crimes recorded (No.) by the police authorities by Geographic location
7. Population density (No./km²) by Place of residence
8. Inequality in the distribution of declared gross income minus assessed personal income tax (IRS) of taxpayers (P80P20 - No.)
9. Inequality in the distribution of declared gross income of tax households (P90P10 - No.)
10. Municipal expenditure on the environment per inhabitant (€/inhab.) by Geographic location
11. Municipal expenditure on performing arts (€) by Geographic location
12. Municipal expenditure on visual arts (€) by Geographic location

13. Municipal expenditure on crafts (€) by Geographic location
14. Municipal expenditure on cultural and creative activities per inhabitant (€) by Geographic location
15. Municipal expenditure on sports activities and facilities per inhabitant (€) by Geographic location
16. Municipal expenditure on interdisciplinary activities (€) by Geographic location
17. Municipal expenditure on audiovisual and multimedia (€) by Geographic location
18. Municipal expenditure on libraries and archives (€) by Geographic location
19. Municipal expenditure on books and periodicals (€) by Geographic location
20. Municipal expenditure on cultural heritage (€) by Geographic location
21. Teachers in non-higher education (No.) by Geographic location
22. Average duration of social security unemployment benefit (Days) by Place of residence
23. Enterprises (No.) by Geographic location
24. Nurses per 1000 inhabitants (No.) by Place of work
25. Exports (€) of goods by Geographic location
26. Average monthly earnings (€) by Geographic location
27. Old-age dependency ratio (No.) by Place of residence
28. Municipal investment in waste management (€) by Geographic location
29. Municipal investment in biodiversity and landscape protection (€) by Geographic location
30. Doctors (No.) by Place of residence
31. Purchasing power per capita by Geographic location
32. Proportion of own revenue in total revenue (%) of municipal councils by Geographic location
33. Ownership registrations of new vehicles per 1000 inhabitants (No.) by Place of residence
34. Net migration (No.) by Place of residence
35. Gross enrolment rate in basic education (%) by Geographic location
36. Gross enrolment rate in secondary education (%) by Geographic location
37. Crime rate (‰) by Geographic location
38. Five-year infant mortality rate (‰) by Place of residence
39. Median value of declared gross income per taxpayer (€) by Geographic location
40. Amount processed of social security family allowance for children and young people (€) by Place of residence

Passive variables (PROFILE):

1. Conventional dwellings (Housing stock - No.) by Geographic location
2. Declared gross income per tax household (€) by Geographic location
3. Area (km^2) of the territorial units by Geographic location
4. Floor area of the rooms completed (m^2) in new constructions for family housing by Geographic location
5. Crude birth rate per thousand inhabitants place of residence

3. Identification of the Dimensions of the Analysis

The identification of the dimensions underlying municipal quality of life was carried out through a Principal Component Analysis (PCA) applied to the previously selected active variables. Before the final extraction of the components, the adequacy of the variables for factor analysis was assessed, and those showing weak statistical adequacy were excluded, namely those with low levels of correlation with the remaining variables. This process made it possible to reduce the initial set of 40 variables to **30 active variables**, significantly improving the overall quality of the analysis.

After this selection stage, the PCA revealed a high overall level of adequacy, reflected in an overall KMO value of 0.802. Although the Kaiser criterion suggested retaining a larger number of components, a solution with **five dimensions** was chosen, so as to ensure the substantive interpretability of the results, avoiding poorly characterised components or components based on a small number of variables. This decision made it possible to obtain dimensions that are conceptually coherent and relevant to the analysis of municipal quality of life.

The substantive interpretation of the principal components was supported by the analysis of the respective loadings, systematised in the file "Interpretacao_Indices_Loadings.csv".

The five dimensions identified were interpreted and named as follows:

- **Dimension 1 – Social Inequality**
  This dimension captures inequality in the distribution of income within municipalities, reflecting internal differences associated with income concentration and socio-economic heterogeneity. Municipalities with high values on this dimension are characterised by greater economic disparities among the resident population.
- **Dimension 2 – Public Services**
  This dimension is associated with the presence and intensity of public intervention in municipalities, reflecting the installed social infrastructure, namely in the fields of education, social protection and public cultural heritage.
- **Dimension 3 – Investment in Culture**
  This dimension conveys the municipal effort in investing in cultural activities, expressing the importance attributed to culture as a component of quality of life and of territorial attractiveness.
- **Dimension 4 – Economic Engine**
  This dimension represents the economic capacity of municipalities, being associated with higher levels of income, purchasing power and economic dynamism. It is a central dimension for distinguishing economically stronger municipalities from those with lower productive capacity.
- **Dimension 5 – Human Capital**
  This dimension reflects the level of human capital development, incorporating indicators related to health and education. Municipalities with high values on this

dimension tend to show better conditions of access to health services and higher levels of schooling.

The dimensions obtained make it possible to synthesise, in a structured and interpretable way, the complexity of quality of life at the municipal level, serving as a basis for the heterogeneity analysis and for the identification of homogeneous groups of municipalities presented in the following section.

The factor scores of the municipalities on the five principal components are recorded in the file Resultados_Indices_PCA.csv and were used as input in the clustering analysis.

Using the principal components as input for clustering makes it possible to reduce redundancy between variables and to ensure that the segmentation of municipalities is based on orthogonal and statistically consistent dimensions.

4. Identification of heterogeneity in the database

The clustering analysis was carried out on the factor scores obtained in the PCA, making it possible to segment the municipalities on the basis of the previously identified latent dimensions.

It made it possible to identify **five distinct groups of municipalities**, reflecting different quality of life profiles. The heterogeneity observed results from the combination of the economic, social, demographic, territorial and institutional dimensions synthesised by the principal components. The clusters were characterised on the basis of the **passive variables (PROFILE)**, using standardised values, so as to identify relative deviations from the overall mean.

The external validation and substantive characterisation of the clusters based on the PROFILE variables were carried out using the file “Resumo_Perfil_Externo.csv”.

The final assignment of the municipalities to the five clusters identified is systematised in the file “Resultados_Clustering_Final.csv”

In turn, the mean characterisation of the clusters on the principal components was supported by the information summarised in the file “Resumo_Perfil_Clusters.csv”.

Cluster 1 – Intermediate-profile municipalities is the group that best represents the national average. The municipalities included in this cluster show values close to the mean on all the profile variables considered, with no significant deviations in economic, demographic or housing terms. This group thus reflects a balanced and relatively homogeneous level of quality of life, serving as a reference for comparison with the other clusters.

Cluster 2 – Metropolitan municipalities brings together municipalities with strong economic dynamism, evidenced by high income levels, but at the same time marked by high housing pressure and weaker demographic dynamics. The combination of these factors suggests dense urban contexts, with greater economic capacity, but also with constraints associated with access to housing and more intensive lifestyles, reflecting specific challenges to quality of life in these territories.

Cluster 3 – Low-density municipalities is characterised by high availability of land and housing per inhabitant, associated with contexts of low population density. These municipalities, however, show unfavourable demographic dynamics, reflected in low birth rates, often associated with ageing population structures. This profile is consistent with territories mostly in the interior of the country, where quality of life is marked by lower urban pressure, but also by limitations in terms of economic and demographic dynamism.

Cluster 4 – Municipalities in urban transition presents an intermediate profile, combining characteristics of urban municipalities with features typical of less dense territories. Although they show some positive demographic dynamics, these municipalities continue to reveal weaknesses in terms of housing supply and average income. This cluster can be interpreted as representative of territories undergoing a process of transformation, lying slightly above the national average in terms of quality of life, but still with structural challenges to be resolved.

Finally, **Cluster 5 – Municipalities with demographic growth** clearly stands out for its positive population dynamics, reflected in above-average birth rates and greater activity associated with housing construction. Although their income levels are still close to or slightly below the national average, these municipalities show consistent signs of population growth and renewal, suggesting a favourable potential for the future evolution of quality of life.

Taken together, the clusters identified show the existence of **differentiated territorial patterns of quality of life**, confirming that the phenomenon analysed is marked by strong spatial heterogeneity. The typology obtained thus allows a structured reading of municipal differences, providing a solid basis for strategic reflection and for decision support in the context of the analysis of territorial quality of life.

5. Conclusion

The aim of this study was to build a **municipal quality of life index**, using unsupervised statistical learning techniques, with a view to identifying latent patterns and classifying Portuguese municipalities into homogeneous profiles. The analysis carried out made it possible to synthesise a broad set of official indicators into interpretable dimensions and to reveal the existence of marked territorial heterogeneity in terms of quality of life.

The results show that municipalities associated with **emerging urban contexts**, characterised by greater demographic dynamism and a more active housing supply, tend to present more favourable quality of life profiles. In contrast, strongly urban municipalities reveal, despite their economic dynamism, significant constraints in terms of housing and demographic dynamics, while low-density municipalities face challenges related to population ageing and lower economic vitality.

One of the main conclusions of the study is the absence of a municipal profile that consistently concentrates high values on all the dimensions analysed. This result shows that municipal quality of life results from **trade-offs between different**

dimensions, with no single or ideal model applicable to all territories being observable. In particular, the results suggest the existence of structural tensions between economic dynamism, housing conditions and demographic dynamics, reflecting challenges currently faced by Portuguese municipalities.

In substantive terms, the index and the typology of municipalities obtained are a useful analytical instrument to support strategic reflection on territorial and quality of life policies. Although the study focuses exclusively on the Portuguese context, the methodology adopted is easily replicable in other territorial settings, in line with the broader objective of comparative analysis of quality of life at the Mediterranean scale.

It should be stressed that the results depend on the selection of variables and on the time period considered, and should not be interpreted as an absolute measure of quality of life.

6. Appendices

6.1 Data preparation

6.1.1 Construction and harmonisation of the database

The files used in this study were downloaded directly from the platform of the Instituto Nacional de Estatística in CSV format. In all the files, the column containing both the code and the name of the municipality was normalised to the designation **“Localização geográfica”** (Geographic location), allowing a common key to exist between the different datasets.

In the R environment, this column was then split into **municipality code** and **municipality name**, using the separator “:”. During this process, inconsistencies were identified, namely municipalities with different codes for the same designation and municipalities with identical names belonging to different regions (Mainland, Azores and Madeira).

To resolve these situations, a **Master Table** was built, containing the code, the name of the municipality and the respective region, identified on the basis of the code prefix. From this table a **unique identifier (ID_UNICO)** was created, allowing the correct aggregation of duplicate records and ensuring an unambiguous correspondence between municipalities in all stages of the analysis.

6.1.2 Normalisation of the variables by population

In order to make the indicators comparable across municipalities of different sizes, the variables were transformed to a **per inhabitant** scale, using the variable “Resident population (No.) by Place of residence”. The variables originally expressed in absolute values were divided by the resident population of each municipality.

The variables already expressed as percentages or per inhabitant did not undergo any additional transformation. In the case of the variables expressed per 1000 inhabitants, only the corresponding scale conversion was performed.

6.1.3 Treatment of missing data

The analysis of missing data was carried out using the **VIM** and **MICE** packages, and a missingness pattern of the **MNAR (Missing Not At Random)** type was identified.

Although the **PMM (Predictive Mean Matching)** method is frequently used in multiple imputation, it was decided not to apply it in this study because of the characteristics of the data. PMM relies on linear regression models, assuming approximately linear relationships between variables, which was not consistently verified in the municipal database analysed.

In contrast, the **CART (Classification and Regression Trees)** algorithm makes it possible to capture non-linear relationships and complex interactions between variables, proving more suitable for dealing with territorial heterogeneity and the structural diversity of municipalities. For this reason, the CART method was considered more appropriate for the treatment of missing data in the present study.

Before imputation, the identification columns were removed, allowing the model to operate exclusively on the numerical variables. A correlation analysis was also carried out, with removal of highly correlated variables, so as to avoid redundancy in the imputation process.

The result of this process is the final database used in the study, recorded in the file “dataset_completo_imputado.csv”.

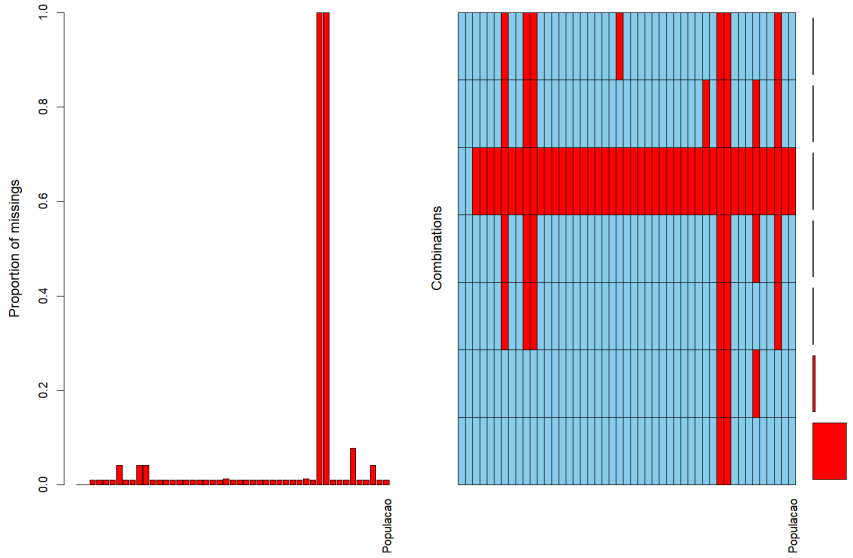


Figure 1 – Proportion of missing data and combination patterns between variables

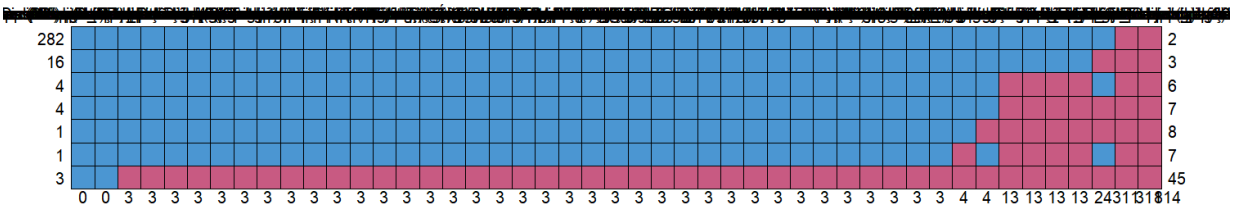


Figure 2 – Missing data pattern before imputation

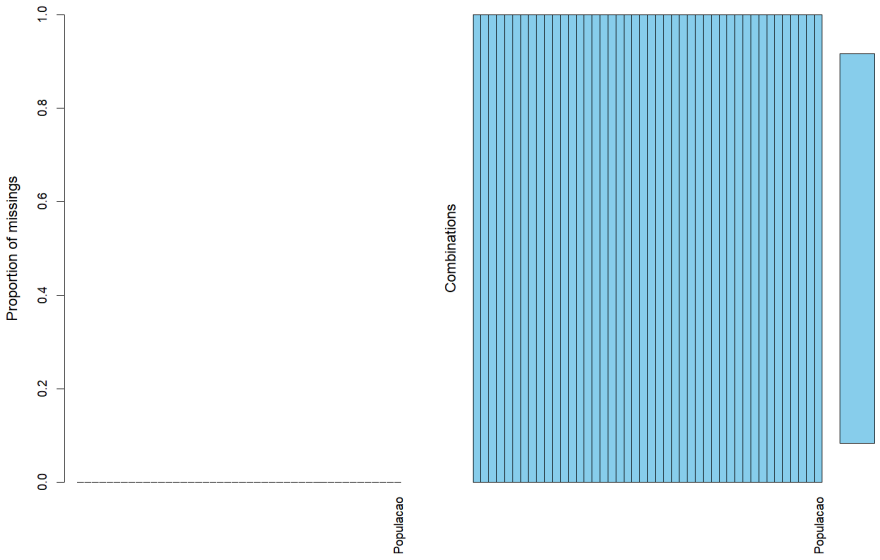


Figure 3 – Data pattern after multiple imputation

6.2 Principal Component Analysis (PCA)

6.2.1 Verification of data adequacy

The adequacy of the data for Principal Component Analysis was assessed using the **Kaiser-Meyer-Olkin (KMO)** test. At an initial stage, the overall KMO value proved acceptable (0.633), and was subsequently improved to 0.802 through the selection of variables with greater factor adequacy.

```
--- TESTE DE ADEQUAÇÃO KMO (Kaiser-Meyer-Olkin) ---
> kmo_res <- KMO(dados_ativos)
> print(kmo_res$MSA) # Imprime o KMO global
[1] 0.8020533
```

Figure 4 - Final overall KMO test

6.2.2 Determination of the number of components

The initial number of components was determined on the basis of the **Kaiser criterion**, and was subsequently adjusted on the basis of the analysis of the interpretability of the components and of the distribution of the factor loadings.

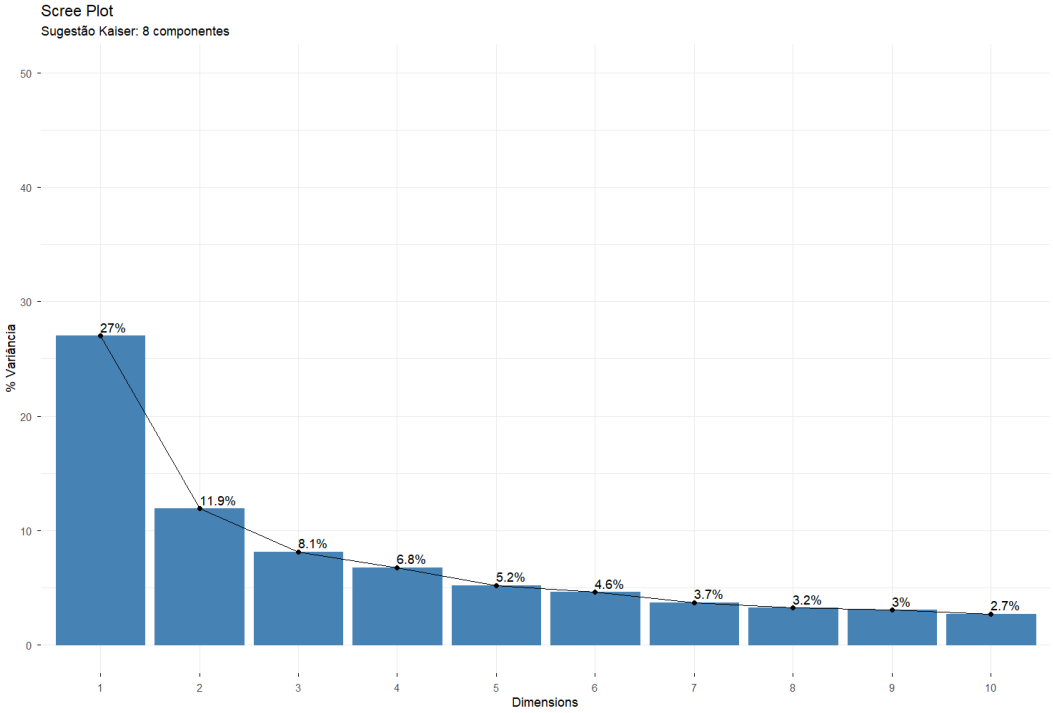


Figure 5 – Scree Plot (Kaiser criterion)

6.2.3 Rotation and Interpretation

The final interpretation of the principal components was carried out through an orthogonal rotation using the **Varimax** method, allowing a clearer and more interpretable structure of the dimensions identified.

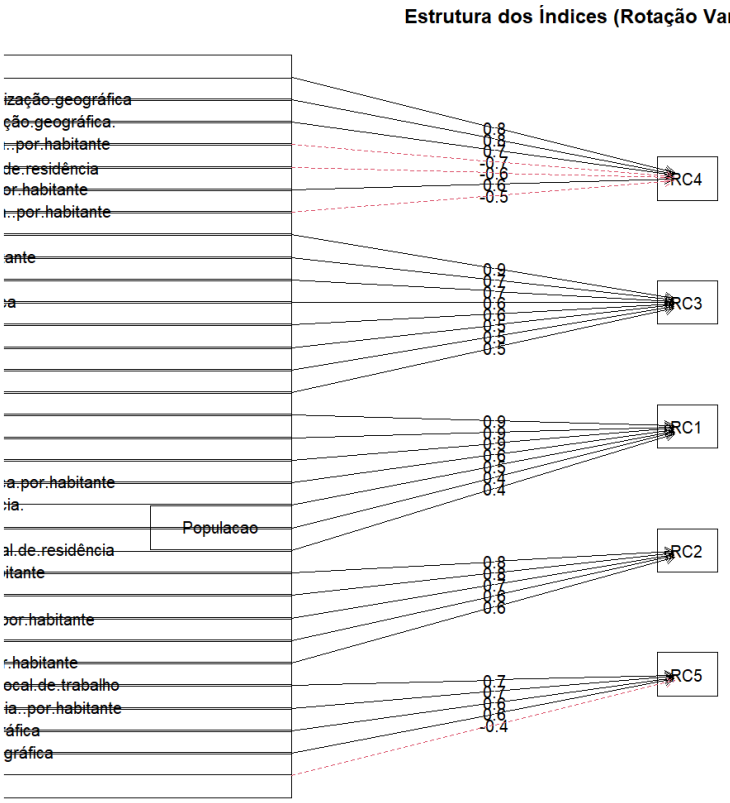


Figure 6 - Final structure of the PCAs

```
                         RC4  RC3  RC1  RC2  RC5
SS loadings             4.45 4.39 4.18 3.27 2.59
Proportion Var          0.14 0.14 0.13 0.10 0.08
Cumulative Var          0.14 0.28 0.41 0.51 0.59
Proportion Explained    0.24 0.23 0.22 0.17 0.14
Cumulative Proportion   0.24 0.47 0.69 0.86 1.00

Mean item complexity =  1.8
Test of the hypothesis that 5 components are sufficient.

The root mean square of the residuals (RMSR) is  0.06
 with the empirical chi square  1175.81  with prob <  6.6e-91

Fit based upon off diagonal values = 0.95>
```

Figure 7 – PCA fit statistics and variance explained by the components

6.3 Clustering

6.3.1 Distance Measure

The distance measure adopted in the clustering analysis was the Euclidean distance. This choice is directly related to the nature of the data used, since clustering was applied to the principal components obtained in the Principal Component Analysis. Given that these components are orthogonal and standardised, it was not necessary to resort to distance measures that make additional corrections for correlation between variables.

In addition, the Euclidean distance has high geometric interpretability, representing the shortest straight line between two profiles in a multidimensional space. This characteristic makes it particularly suitable for measuring the overall similarity between municipalities when the scales of the data are already normalised, allowing a clear and intuitive reading of the proximities and differences between territories.

6.3.2 Choice of the Clustering Approach

The clustering approach adopted followed a **hybrid** strategy, combining hierarchical and partitional methods. In an exploratory phase, **hierarchical clustering with Ward's method** was applied, which minimises within-cluster variance and generates a **dendrogram**, facilitating the visualisation of the hierarchical structure of the municipalities. This graphical representation made it possible to identify possible cut-off points and provided a first indication of the appropriate number of clusters to consider.

Based on the results obtained in the exploratory phase, the consolidation phase was carried out by applying the **K-Means** algorithm, using the previously defined number of clusters. This approach makes it possible to refine the initial partition and obtain a more stable and interpretable solution. To increase the robustness of the final solution, a large number of initialisations was used (nstart = 25), allowing the algorithm to evaluate different initial configurations and select the one that minimises the within-cluster sum of squares.

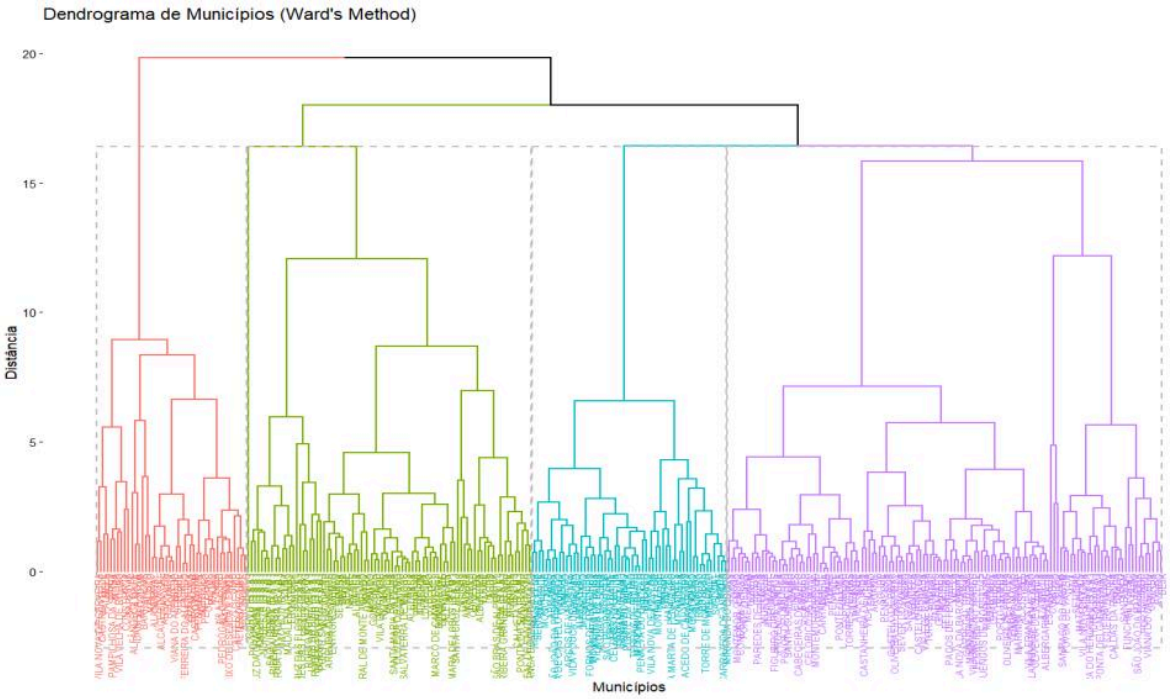


Figure 8 - Dendrogram of Municipalities

6.3.3 Validation of the choice of the number of clusters

The validation of the choice of the final number of clusters was supported by complementary graphical criteria, namely the **Elbow** and **Silhouette** methods. The Elbow method made it possible to assess the evolution of within-cluster variability as a function of the number of clusters, while the Silhouette criterion provided a measure of the quality of the separation between groups.

In addition, the **silhouette plot** of the final solution was analysed, making it possible to assess the consistency of the assignment of municipalities to their respective clusters and to confirm the adequacy of the partition obtained. Taken together, these criteria corroborated the choice of a solution with **five clusters**, considered balanced in terms of statistical quality and substantive interpretability.

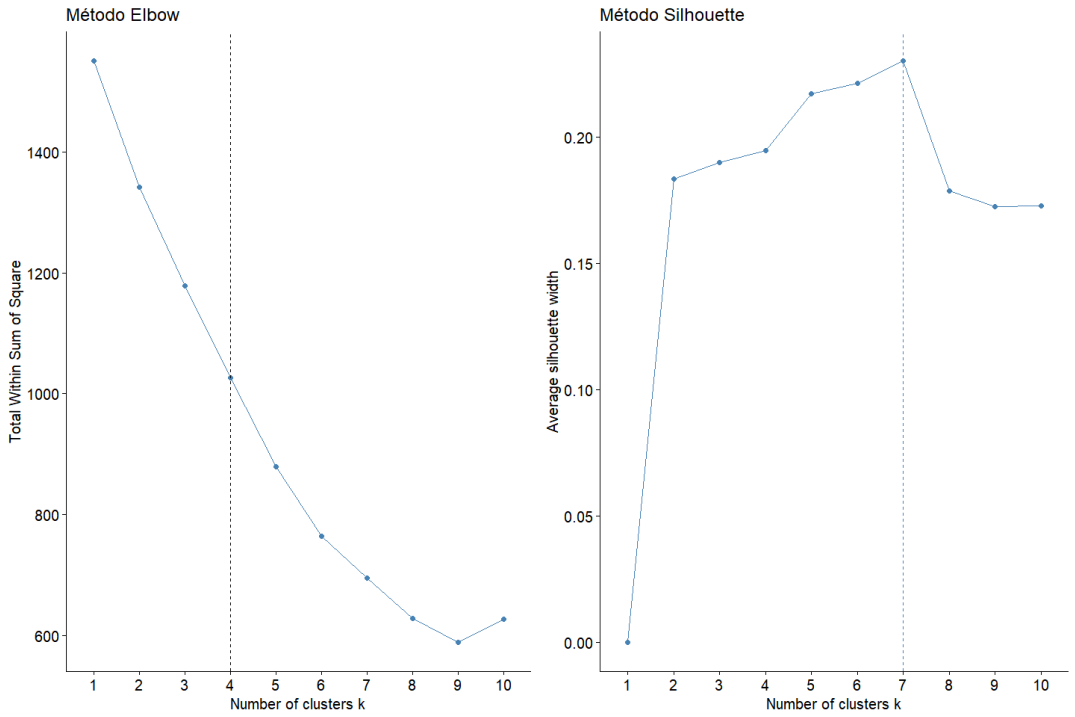


Figure 9 - Elbow and Silhouette methods

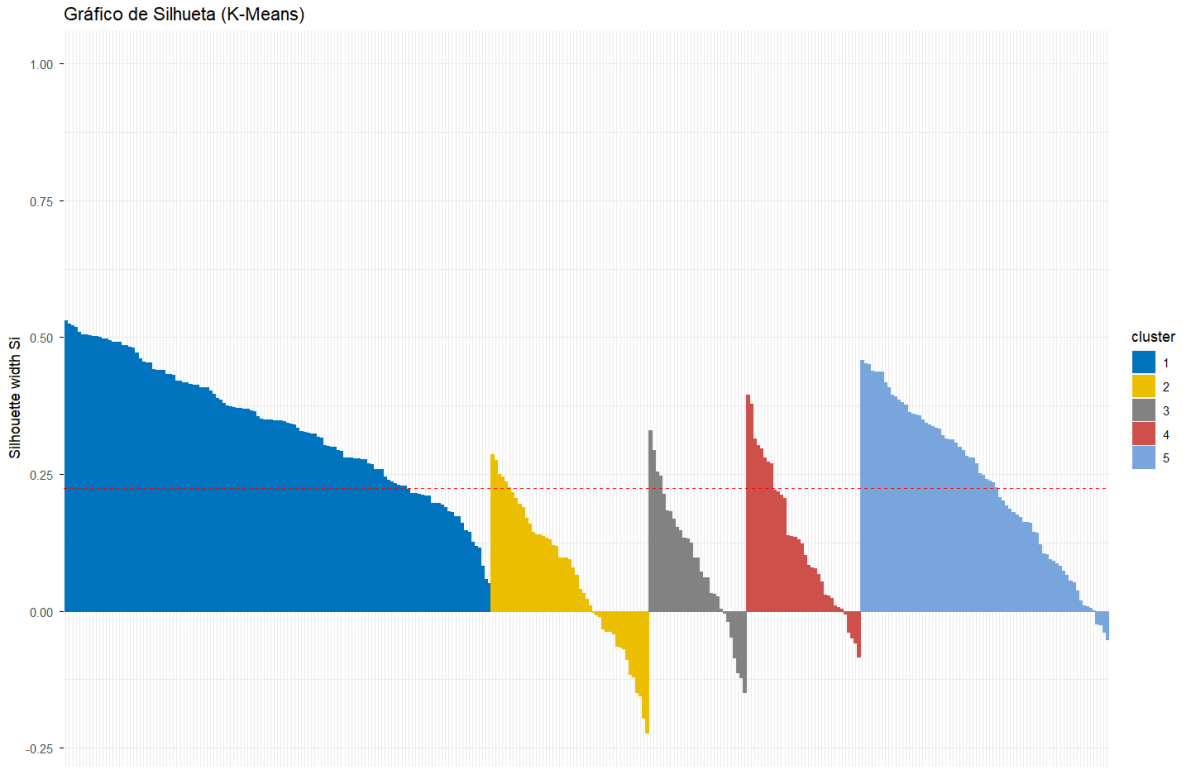


Figure 10 - Silhouette plot of the cluster distribution

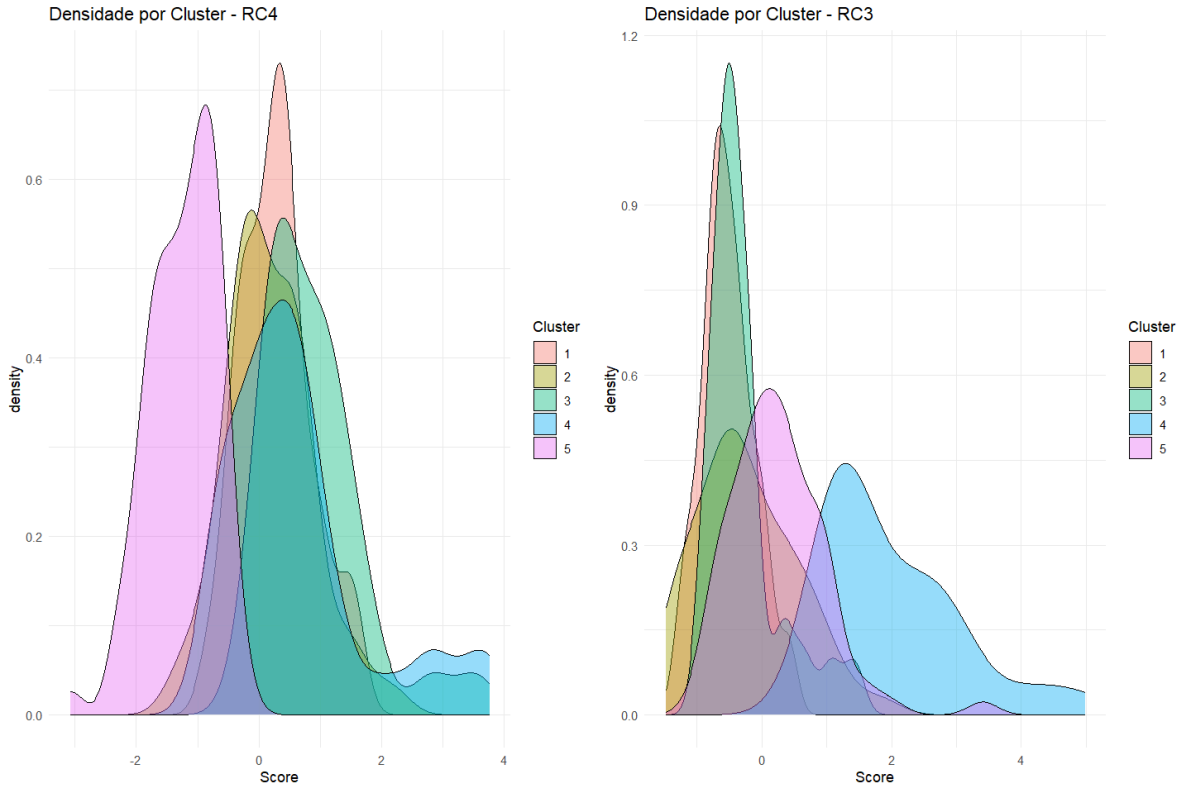


Figure 11 - Density distribution of the RC4 and RC3 component scores by cluster

6.3.4 Representation of the clusters in the principal component space

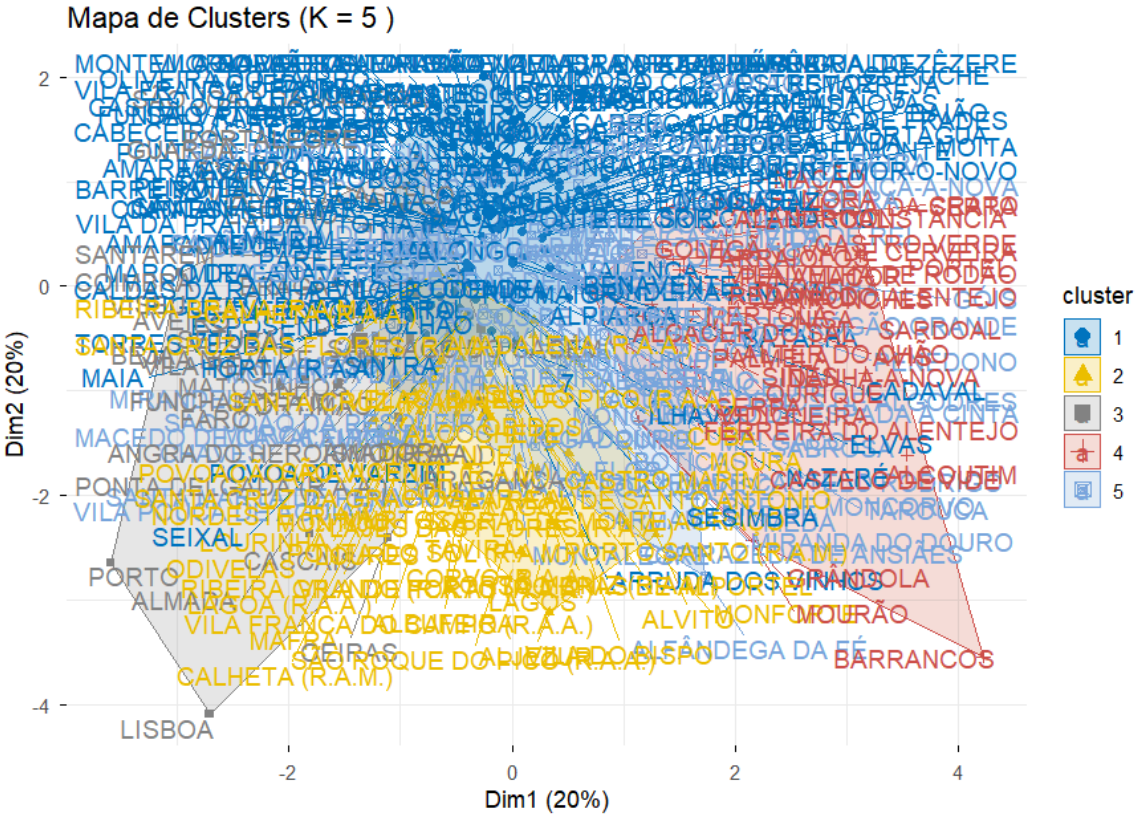


Figure 12 - Cluster Map (K = 5)

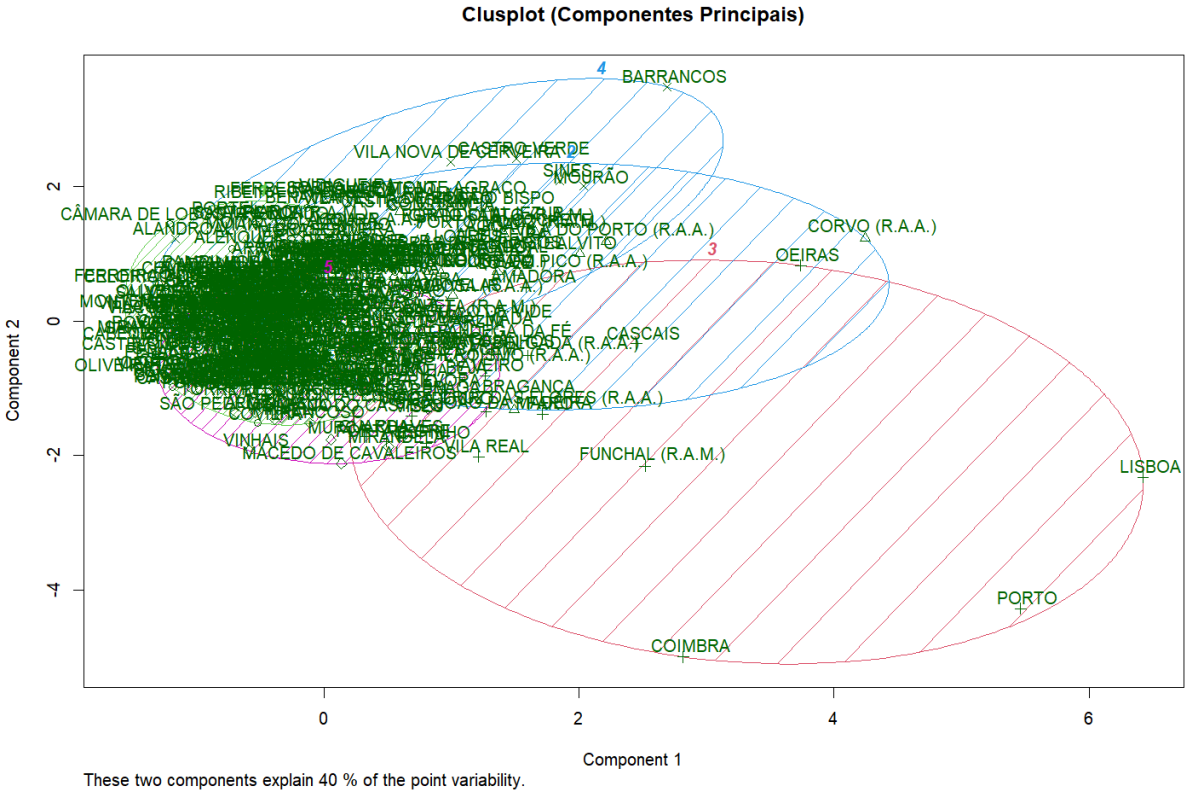


Figure 13 – Clusplot (Principal Components)

6.3.5 Mean profile of the clusters on the principal components

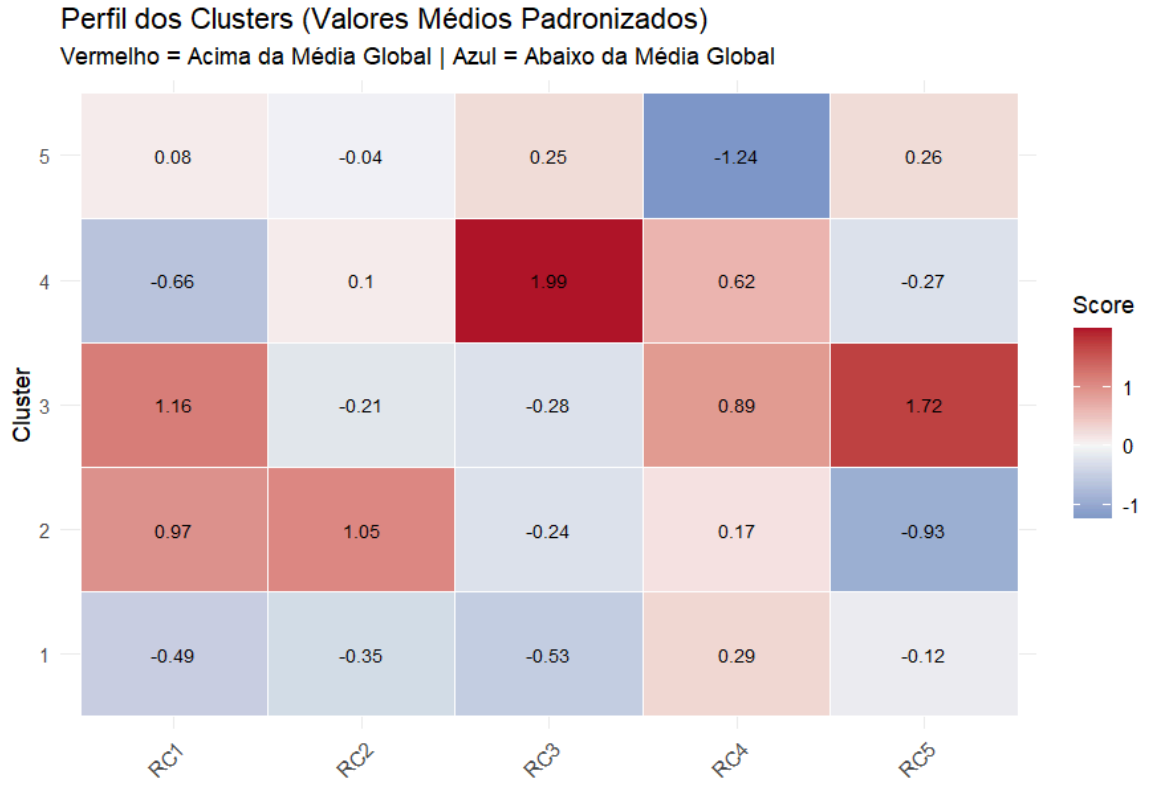


Figure 14 – Cluster Profile (Standardised Mean Values)

6.3.6 Extended profile of the clusters based on the validation variables

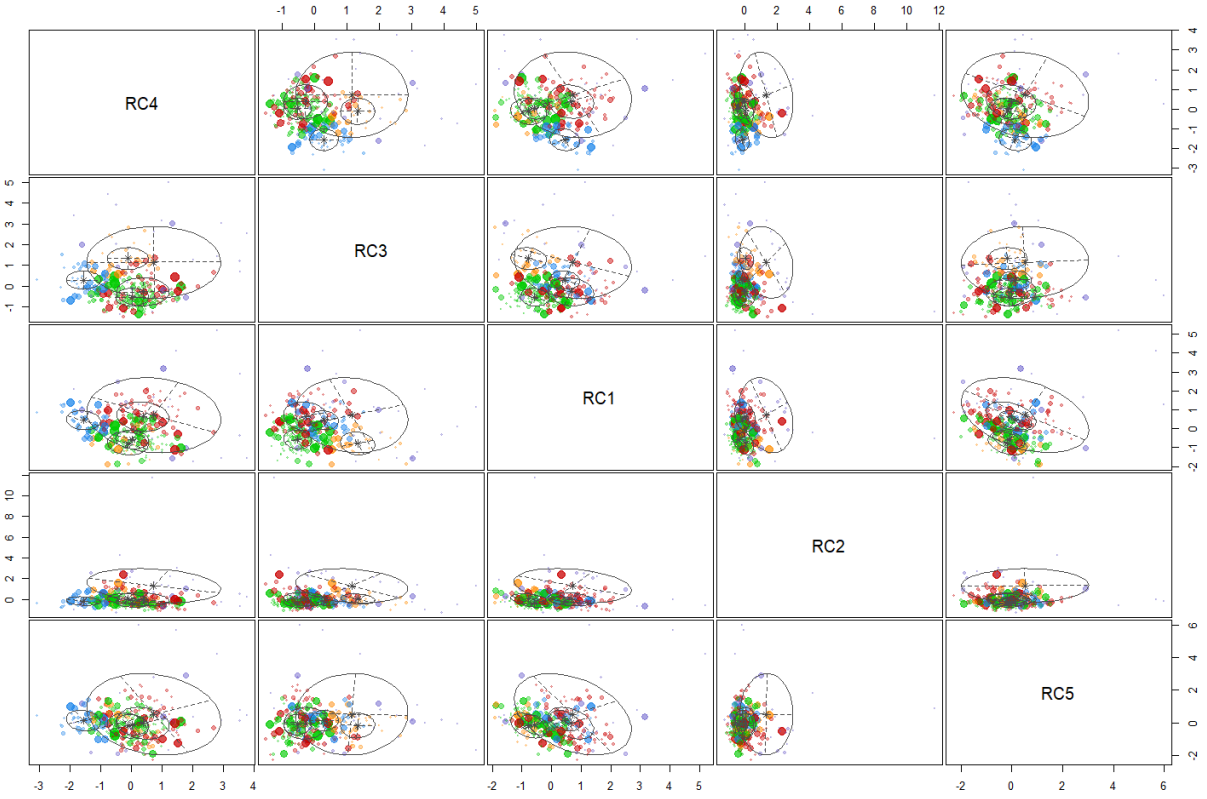


Figure 15 - Matrix of bivariate scatterplots of the principal components

6.4 Databases and Intermediate Results

- dataset_completo.csv
  - ✓ Initial aggregated database
- dataset_completo_imputado.csv
  - ✓ Final database after treatment of missing data
- Interpretacao_Indices_Loadings.csv
  - ✓ Loadings and interpretation of the principal components
- Resultados_Indices_PCA.csv
  - ✓ Factor scores of the municipalities
- Resultados_Clustering_Final.csv
  - ✓ Final assignment of the municipalities to the clusters
- Resumo_Perfil_Clusters.csv
  - ✓ Mean profiles of the clusters on the principal components
- Resumo_Perfil_Externo.csv
  - ✓ External validation variables of the clusters